

TrendMart International

Global E-Commerce Platform Migration

Technical Architecture Document

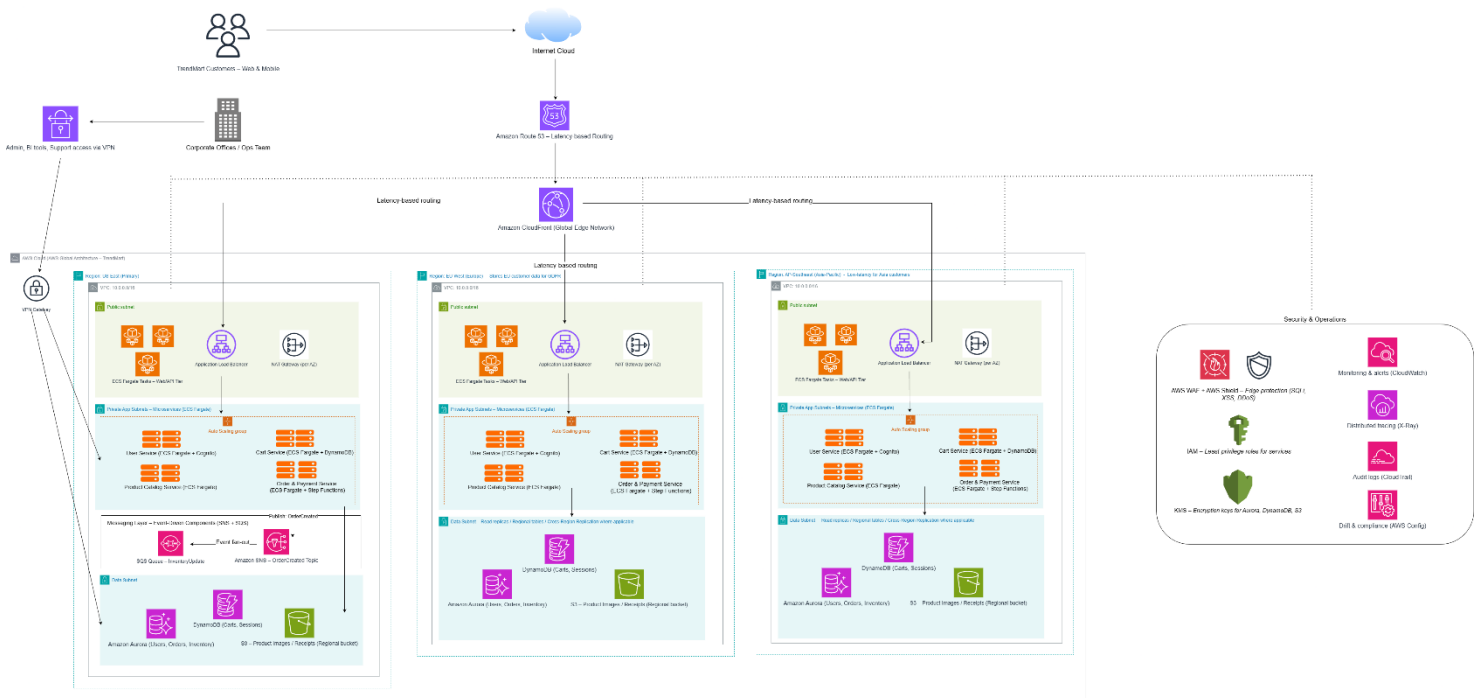
FINAL EXAM USE-CASE

Student Name: **Harish Nandha Kumar**

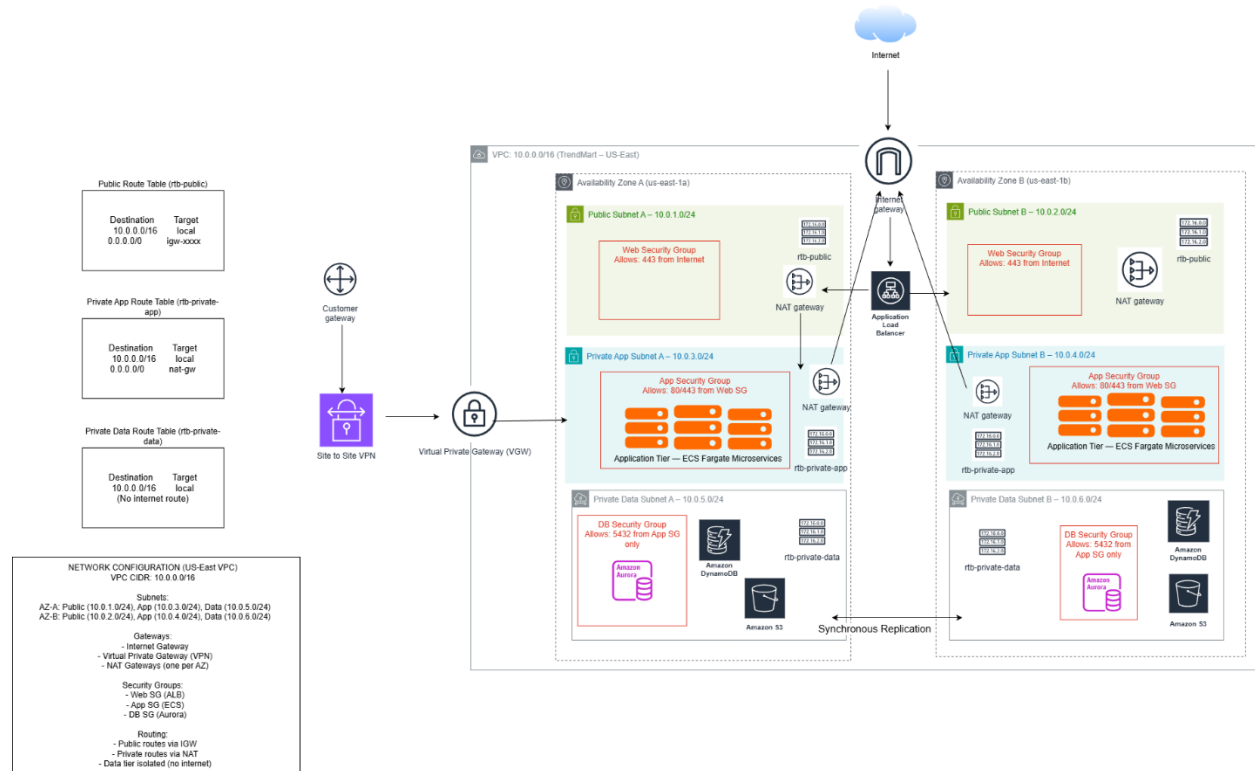
Date: **December 4th, 2025**



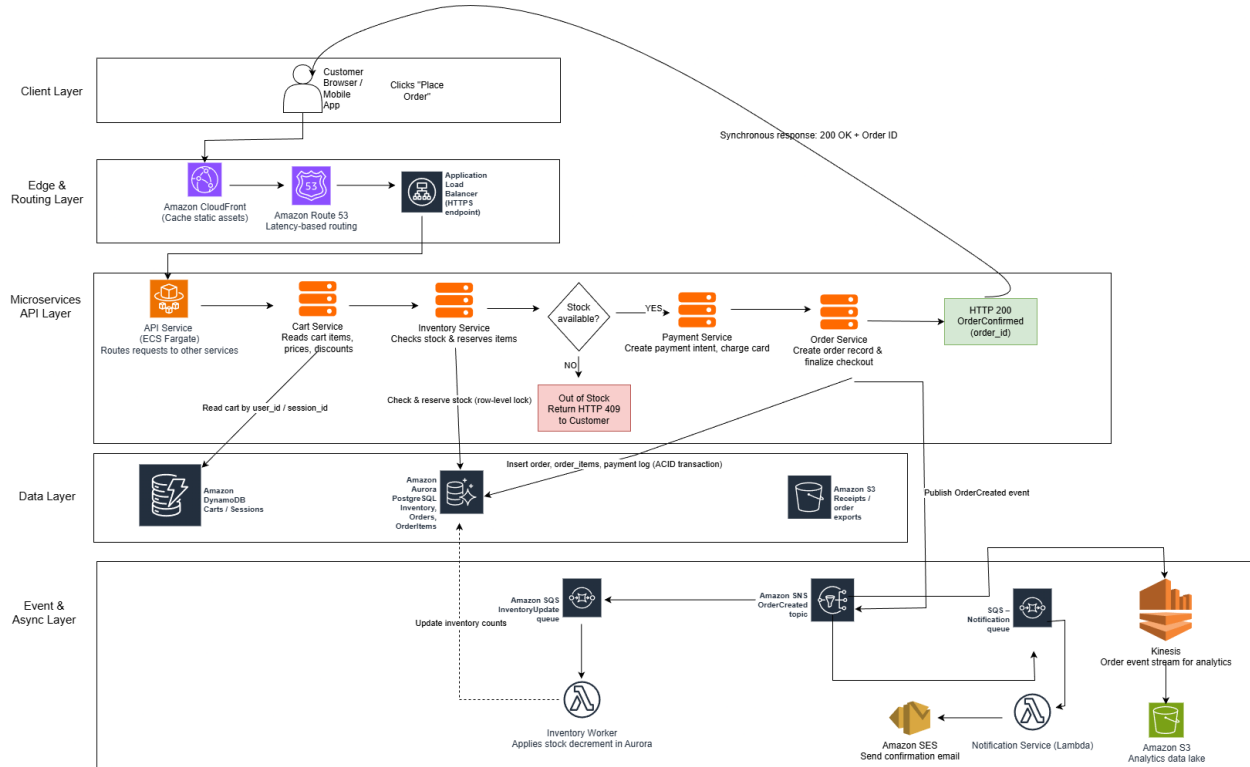
(1). OVERALL SYSTEM ARCHITECTURE:



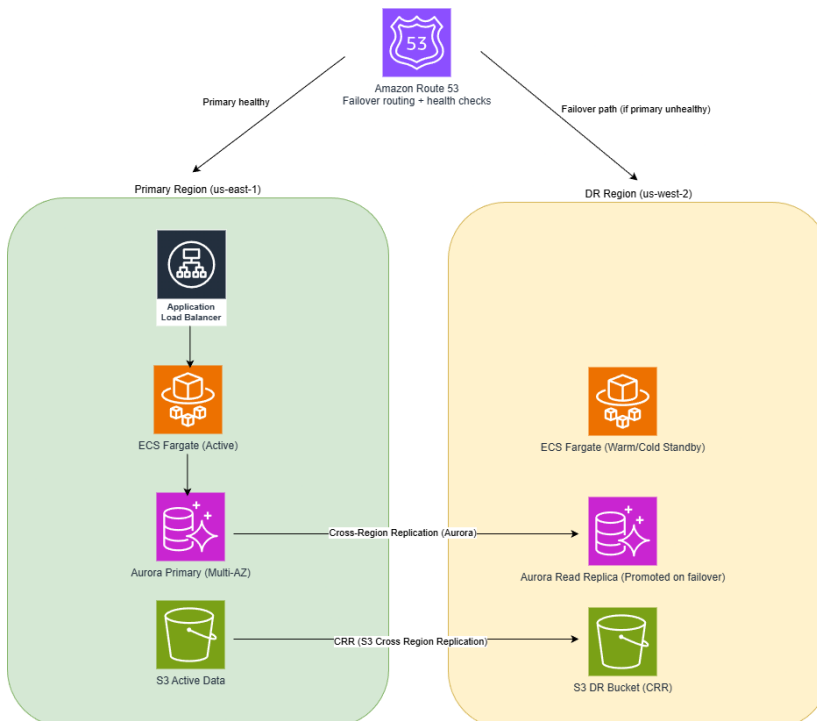
(2). NETWORK DESIGN AND SECURITY ZONES:



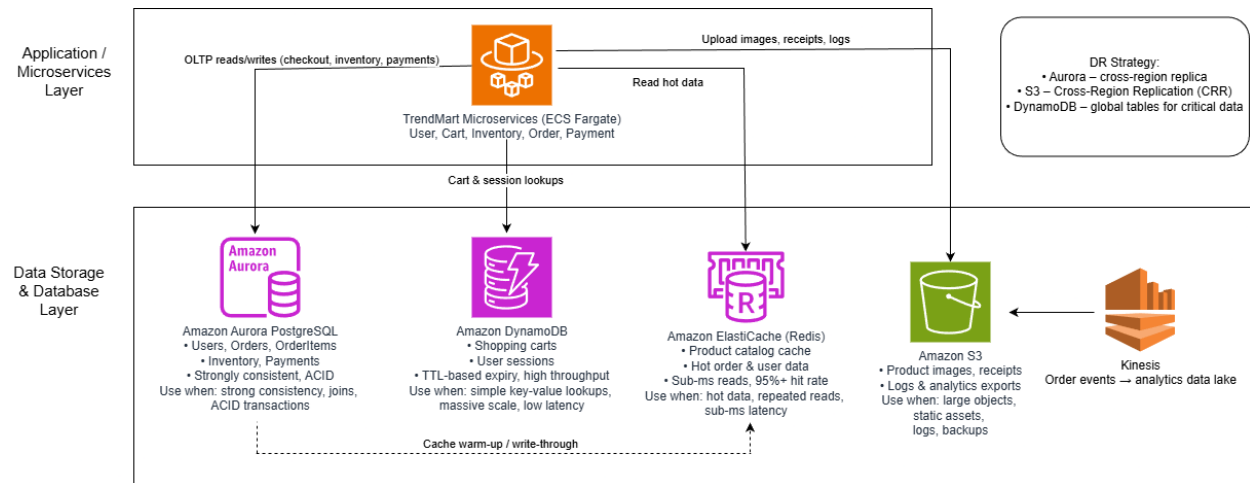
(3). HOW AN ORDER FLOWS THROUGH THE SYSTEM (DATA FLOW DIAGRAM):



(4). DISASTER RECOVERY DESIGN DIAGRAM:



(5). DATA STORAGE AND DATABASE ARCHITECTURE DIAGRAM:



CURRENT PROBLEMS:

TrendMart faces many problems due to its 15 years old outdated monolithic system as it still uses more than 200 physical servers but still cannot meet its modern e-commerce demands. The existing monolithic system pushes the entire application to scale as one whole unit as this causes failures during peak shopping events such as Black Friday where the network traffic spikes 100x more than the normal load. In addition, TrendMart's permanently over-provisions infrastructure is a major disadvantage as if any failure occurs, then it takes more months to procure the replacement hardware. So, all this time, the services is not provided to its full potential and it may lead millions in lost revenue. Further, the current single 30TB database suffers heavy read/write contention which leads to slowing down of product browsing and inventory inconsistencies. Moreover, the data center hosting all the data is inside US, so it causes latency issues to the international customers. Further, they face compliance issues with EU as the European customer data stored outside the EU has already resulted in GDPR fines. Finally, the application's code is tightly coupled due to its monolithic system and the lack of CI/CD pipelines makes the system updates to occur slowly and time-consuming. Thus, these issues affect the TrendMart from providing faster services and makes it outdated relative to its competitors.

PROPOSED SOLUTION:

The proposed solution adopts the microservices-based design that utilizes the AWS managed services as it aims to reduce the TrendMart's operational burden with their limited cloud-experienced team. In this solution, the AWS ECS with AWS fargate is used to compute the

workloads as it helps to remove the server management and provides automatic scaling based on load for handling 1,000,000 of peak users. Then, the AWS CloudFront helps to deliver static contents from edge locations globally as it further leads to reduction in page load times to sub-200 milliseconds. The Elastic load balancers along with the health checks enabled is utilized to ensure the reliability. Further, the application runs in multiple AWS Regions (US-East, EU-West, AP-Southeast) and utilizes Route 53 to provide latency-based routing for ensuring global performance and data residency. Further, in the data tier, AWS Aurora is used for transactional workloads and AWS DynamoDB is utilized to provide high-speed scalable cart service and session data. In addition, the AWS SQS helps to absorb traffic spikes in order to prevent cascading failures during flash sales. Thus, this architecture is designed to ensure reliability and aims to improve user experience while also following the compliance requirements.

MIGRATION PLAN:

TrendMart is using a gradual migration strategy to avoid downtime. Firstly, the foundational services such as AWS Organizations, IAM, CloudTrail and AWS VPC are established to ensure a secure and governed environment. Then, the initial microservices such as Product Catalog is taken from the monolith and deployed to ECS Fargate. In addition to it, the data is migrated to Aurora using AWS Database Migration Service only with minimal disruption. Further, the Route 53 helps to direct a small percentage of traffic to the new service for real-world validation before full cutover. Similarly, Other microservices such as User Management, Cart, Orders, Inventory, Payment services are migrated to the newer one over next several months. Until then, the legacy system remains operational before all other operations are moved there completely. Meanwhile, the existing internal engineering team receives the training and hands-on workshops to learn to adapt to the AWS cloud computing environment. Once, the entire transition is done, the legacy servers are retired in order to eliminate infrastructure costs.

SERVICE DESIGN:

The existing monolith will be decomposed into five core microservices, each mapped to an AWS-managed service to keep operations simple and each service exposes REST APIs through AWS API Gateway which helps to ensure consistent routing, throttling, and security across all microservices:

(i). User Service: The User service runs on AWS ECS Fargate and utilizes AWS Cognito to provide User Authentication services as it helps to eliminate the need for custom identity code. Here, the

Cognito manages to do the signups operations, enable Multi-factor Authentications, Password resets and handle the profile logic services.

(ii). Product Catalog Service: The AWS Aurora along with CloudFront and S3 helps to store the product information and helps to serve the product images for the catalog. In addition, the AWS ElastiCache is utilized to reduce the database load by caching the Frequently accessed items.

(iii). Shopping Cart Service: This service uses the DynamoDB to enable the millisecond reads/writes and seamless scaling during peak loads. Also, the DynamoDB TTL automatically cleans abandoned carts when those cart operations no longer the servers.

(iv). Inventory Service: Here, the Aurora is utilized for inventory management services and the AWS SQS is utilized to manage the high-volume inventory updates and further helps to prevent the overselling during peak load by processing updates in order.

(v). Order & Payment Service: The AWS Step Functions is utilized to co-ordinate complex workflows in order ensure reliability when the external payment gateways slow down. Further, the Aurora is used to store the finalized records in order.

DATA STRATEGY:

TrendMart adopts to a hybrid data model to match data types with the most suitable AWS-managed service. Firstly, the Aurora (PostgreSQL) supports the relational data such as users, orders, and inventory along with Multi-AZ Failover to ensure reliability for these transactional data. Secondly, the DynamoDB helps to handle the high-volume, low-latency operations such as shopping carts and sessions without the need for provisioning or manual scaling. Further, AWS S3 helps to store the static assets such as product images and order receipts and utilizes the S3 lifecycle cost optimization and CloudFront integration. Moreover, to comply with GDPR, the European customer data is stored only in the EU-West region as the data is enforced via application routing and AWS IAM policies. Finally, the Regular backups, point-in-time recovery, and cross-Region replication features allows us to ensure high availability and make the environment ready to handle the disaster situations.

SECURITY PLAN:

The AWS IAM provides the IAM roles that grants least-privilege access to the users and helps to eliminate the long-loved credentials. Further, AWS WAF allows us to block the SQL injection, XSS and bots attacks at the edge. Then, the AWS Shield Standard provides the automatic DDoS protection on top of the existing protection. Moreover, the KMS encryption ensures to protect

the data at rest across Aurora, DynamoDB, and S3 by encrypting them. Also, the TLS ensures to encrypt and protect the data in transit. Apart from this, The AWS CloudTrail and AWS Config provides continuous compliance monitoring and gives alerts on misconfigurations or when an unexpected activity is detected. Further, GDPR compliance is achieved by restricting European personal data to the EU region and implementing data subject rights via API-driven exports and deletions. Finally, AWS Secret Manager helps to store the database passwords and API keys as it enables us to securely store those sensitive data and also allows for automatic rotation without downtime.

OPERATIONS PLAN:

The AWS CloudWatch enables us to monitor the services health, provide the auto-scaling events, monitor the database load and gives the error rates accurately. In addition, the CloudWatch Alarms notify the teams when the thresholds are exceeded while using the services. Further, the AWS X-Ray enables distributed tracing to identify slow microservice interactions. Also, the AWS OpenSearch Service stores the logs for troubleshooting via CloudWatch Log Subscriptions. Further, the AWS CodePipeline along with CodeDeploy helps to ensure safe and automated releases with instant rollback through the Blue/green deployments. The Aurora Multi-AZ, DynamoDB Global Tables and S3 Cross-region replication used to achieve low RTO/RPO objectives during the disaster recovery operations. Finally, the cost optimization is maintained through the AWS Trusted Advisor, Cost Explorer reports, and Savings Plans for predictable workloads. All these helps to ensure the monthly cost well remains below the \$300K target while maintaining the services available globally and further supports for global expansion.